



ASPERGILLOSIS

Introduction

- Aspergillosis is a disease, usually of the respiratory system, of chickens, turkeys and usually affects the lower pulmonary system with granulomatous lesions in air sacs and lungs.
- Also known as brooder pneumonia, fungal or mycotic pneumonia, pneumomycosis and bronchomycosis
- Acute aspergillosis usually affects young birds and is characterized by high morbidity and mortality
- Chronic aspergillosis occurs in adult birds, especially turkey breeders and is less common

Etiology

- Genus *Aspergillus* is composed of approximately 180 species
- *Aspergillus fumigatus* is the principal agent causing aspergillosis in poultry
- *Aspergillus flavus* is less common
- Other rarely isolated species include *A. terreus*, *A. glaucus*, *A. nidulans*, *A. niger*, *A. nigrescens*, and other

Etiology

- Fungal hyphae are 4–12 μm in diameter and bear conidiophores producing conidia (spores) 2–6 μm in diameter that are easily spread in air
- *Aspergillus* is among the most common mycotoxigenic genera

Host

- Newly hatched turkeys, chickens and ducks are highly susceptible to infection but disease also occurs frequently in neonates of other avian species

Transmission

- By inhalation of spores
- Aspergillosis can be acquired *in ovo*
- Infected eggs that are opened during incubation or hatching, releasing large numbers of spores and contaminating hatchmates

Pathogenesis

- Production of conidia (asexual spores) that are easily dispersed into the air → inhalation of airborne conidia → come in contact with on conjunctival, nasal, tracheal, and air sac epithelium → phagocytosis and killing of conidia by macrophages → macrophages also recruits heterophils at site of infection → Conidia that evade macrophage killing and germinate → invade blood vessel and disseminated to brain, pericardium, bone marrow, kidney, and other tissues →

Pathogenesis

- → marked infiltration of heterophils to destroy hyphae → secretion of proteases enzyme by *A. fumigatus* and other enzymes of heterophils → induced coagulative necrosis of surrounding tissue → formation of granuloma
- In some cases, exudate that becomes lodged in the trachea or syrinx, producing acute asphyxiation and death
- In chronic disease, especially in turkeys, large granuloma obstruct pulmonary blood flow and this causes right-ventricular dilatation and ascites.

Clinical Signs

- Two forms of aspergillosis, acute and chronic
- Biphasic mortality pattern
- In the first 1–3 weeks of age - 5–50% mortality due to acute respiratory infection
 - Within the first 3–5 days infected neonates become dyspnoeic, polypnoeic, with open-mouthed breathing (gasps) due to progressive airway obstruction and died

Clinical Signs

- Survivors develop chronic disease with up to 5% mortality
- Survivors may become lethargic and stunted, develop conjunctival swelling and blindness and exhibit torticollis and other central nervous system abnormalities.
- A common feature of dyspnoea associated with aspergillosis is the lack of rales or other respiratory noises

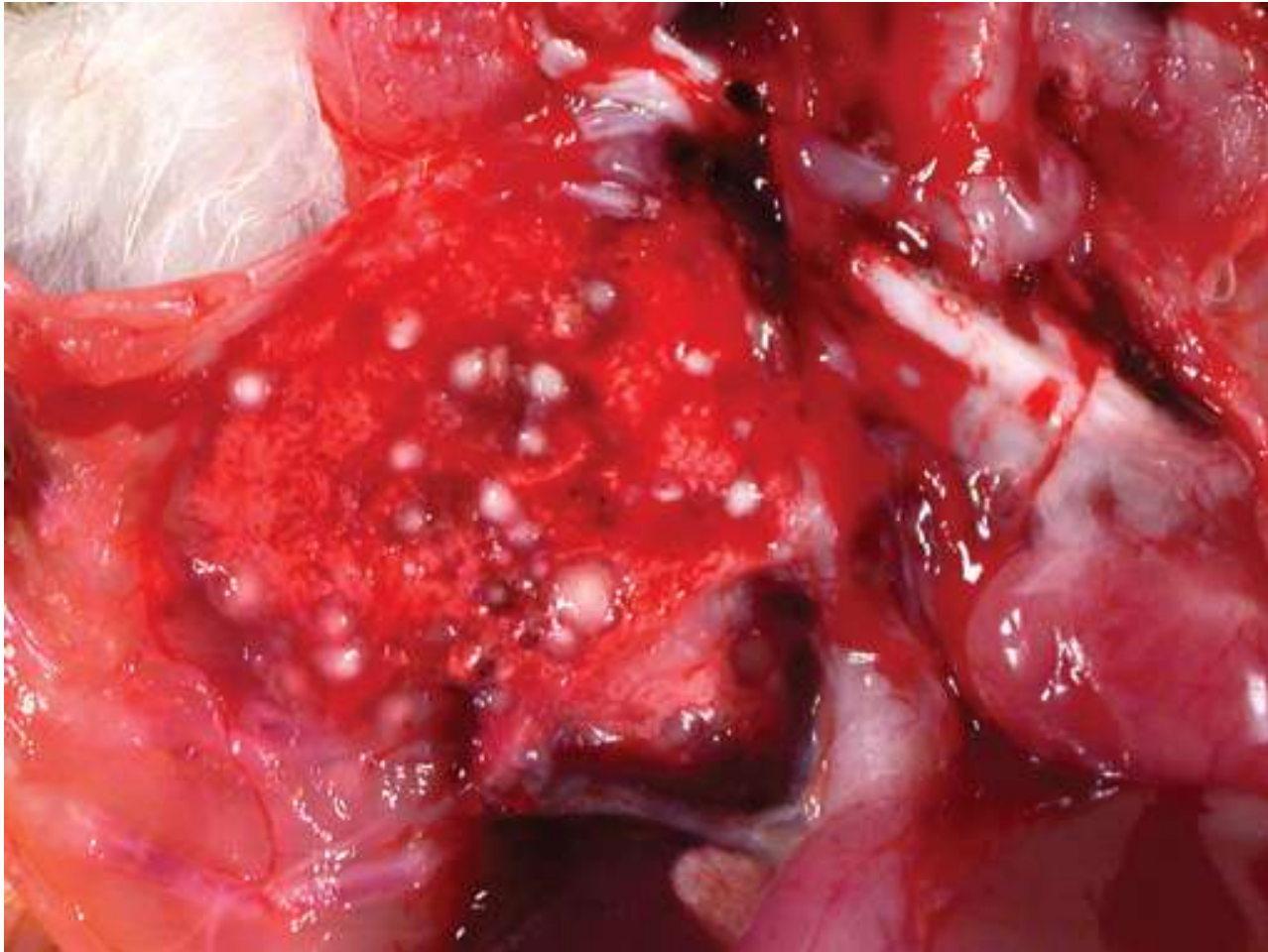
Pathology – Macroscopic

- Lesions are found in trachea, bronchi, lungs and air sacs. Occasionally lesions are seen in the viscera, eye and brain
- Solitary or disseminated white to yellow, discrete, irregular nodules often surrounded by a rim of hyperemia and hemorrhage
- Caseous plaques in air sacs are common in birds
- The lesions that develop in the air sacs are often **dark green-brown** because of the development of fungal conidiophores, which are pigmented

Pathology – Macroscopic



Pathology – Macroscopic



Turkey poult, 2 weeks. Numerous caseous nodules in the lung are characteristic of respiratory aspergillosis (“brooder pneumonia”)

Pathology – Macroscopic



Broiler breeder, 61 weeks. Caseous exudate in the syrinx was caused by aspergillosis

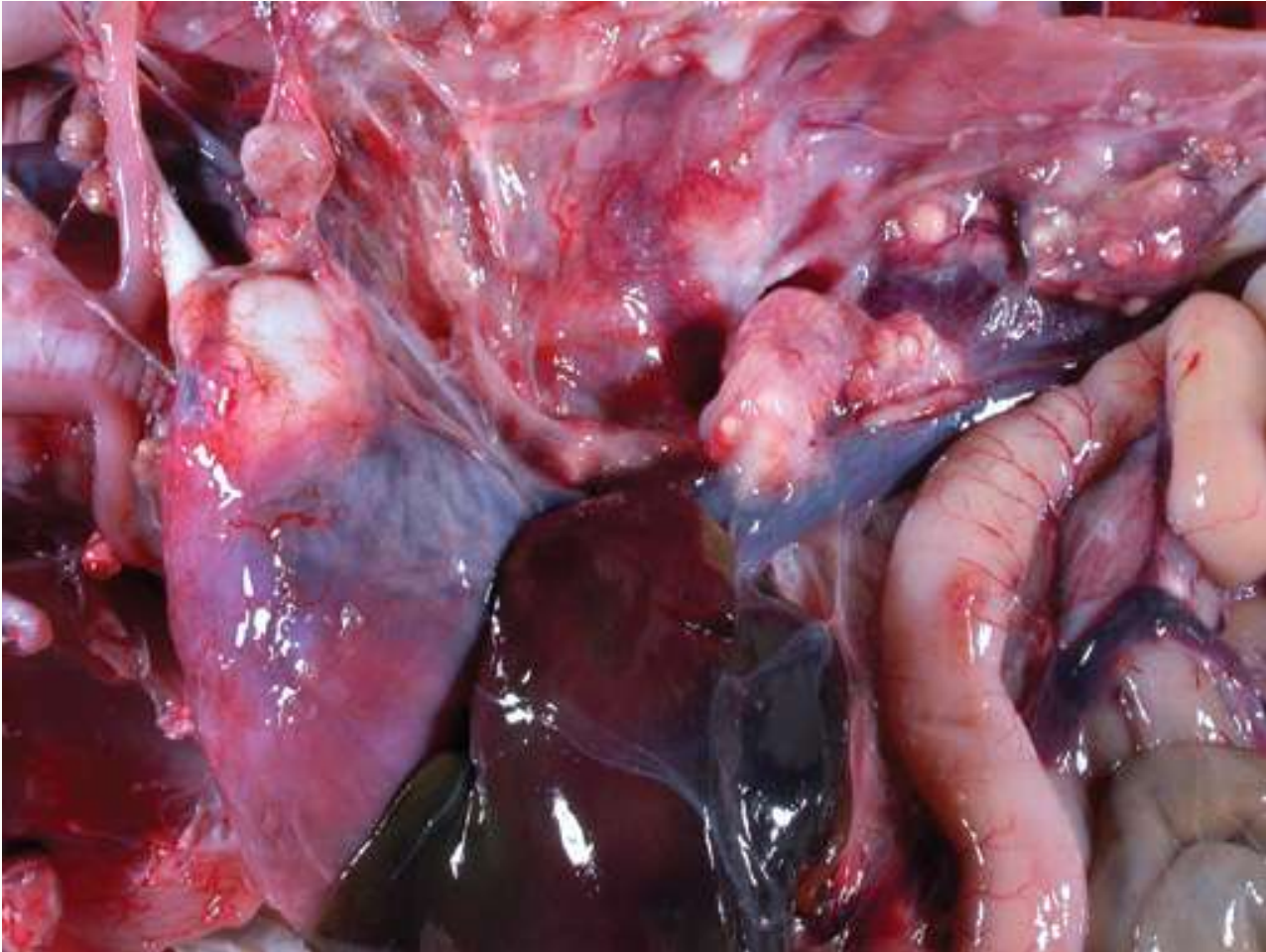
Pathology – Macroscopic



Pathology – Macroscopic



Pathology – Macroscopic



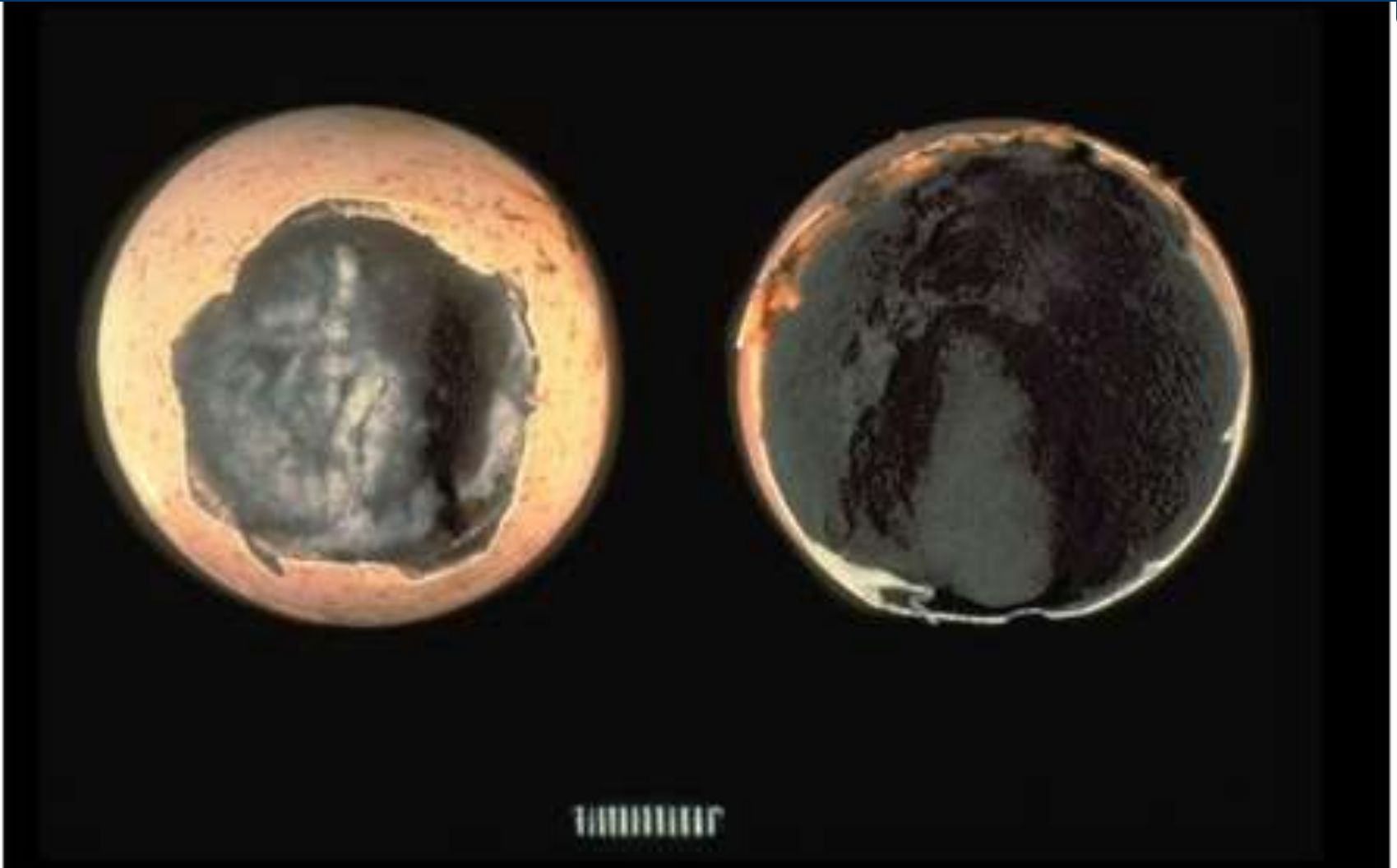
Turkey, 8 weeks. Air sacs containing numerous spherical nodules are granulomas caused by *Aspergillus flavus*

Pathology – Macroscopic



Ocular aspergillosis - This form is characterized by extensive keratoconjunctivitis

Pathology – Macroscopic

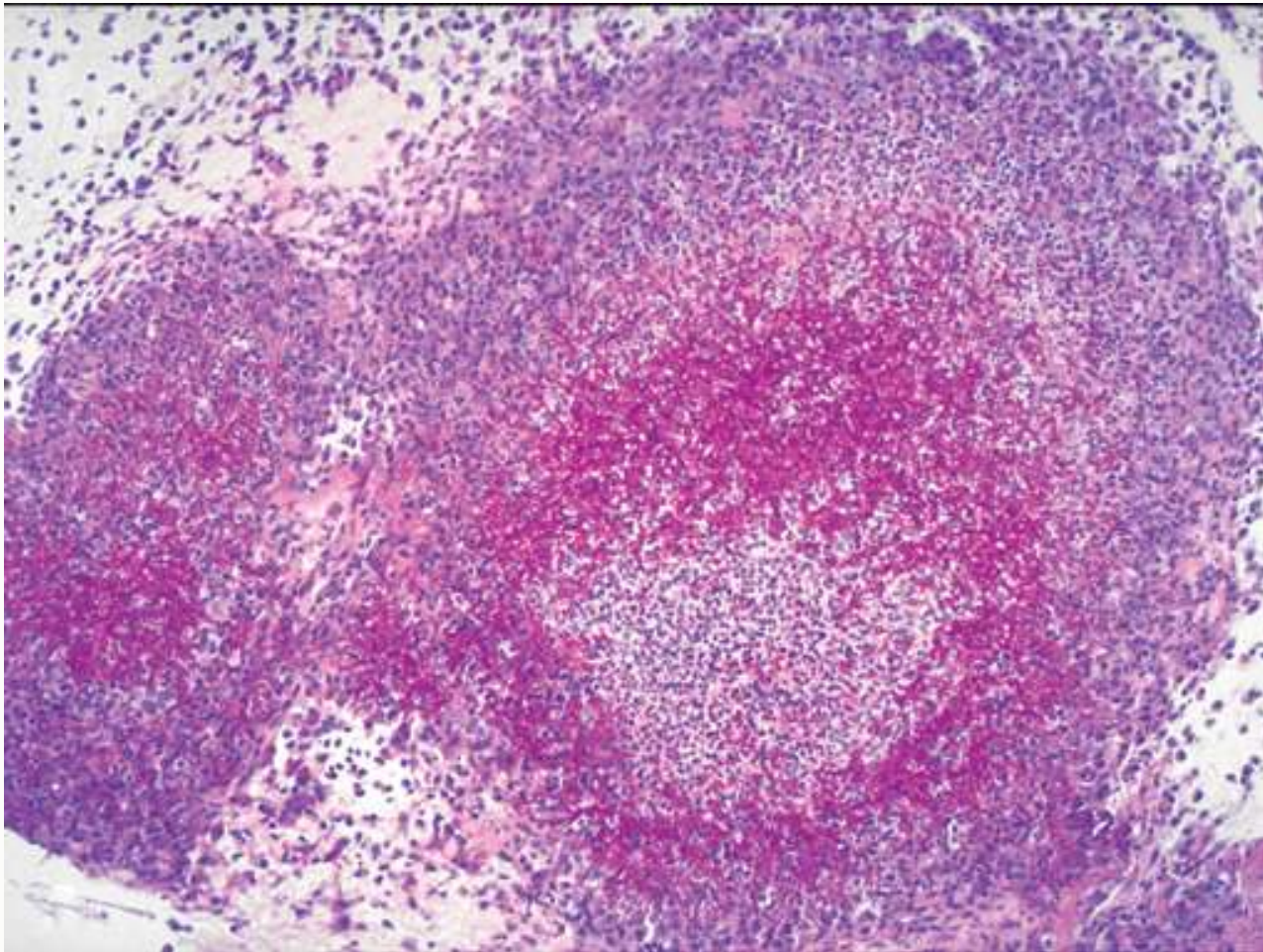


Egg membrane (air cell) mycosis, Aspergillosis

Pathology – Microscopic

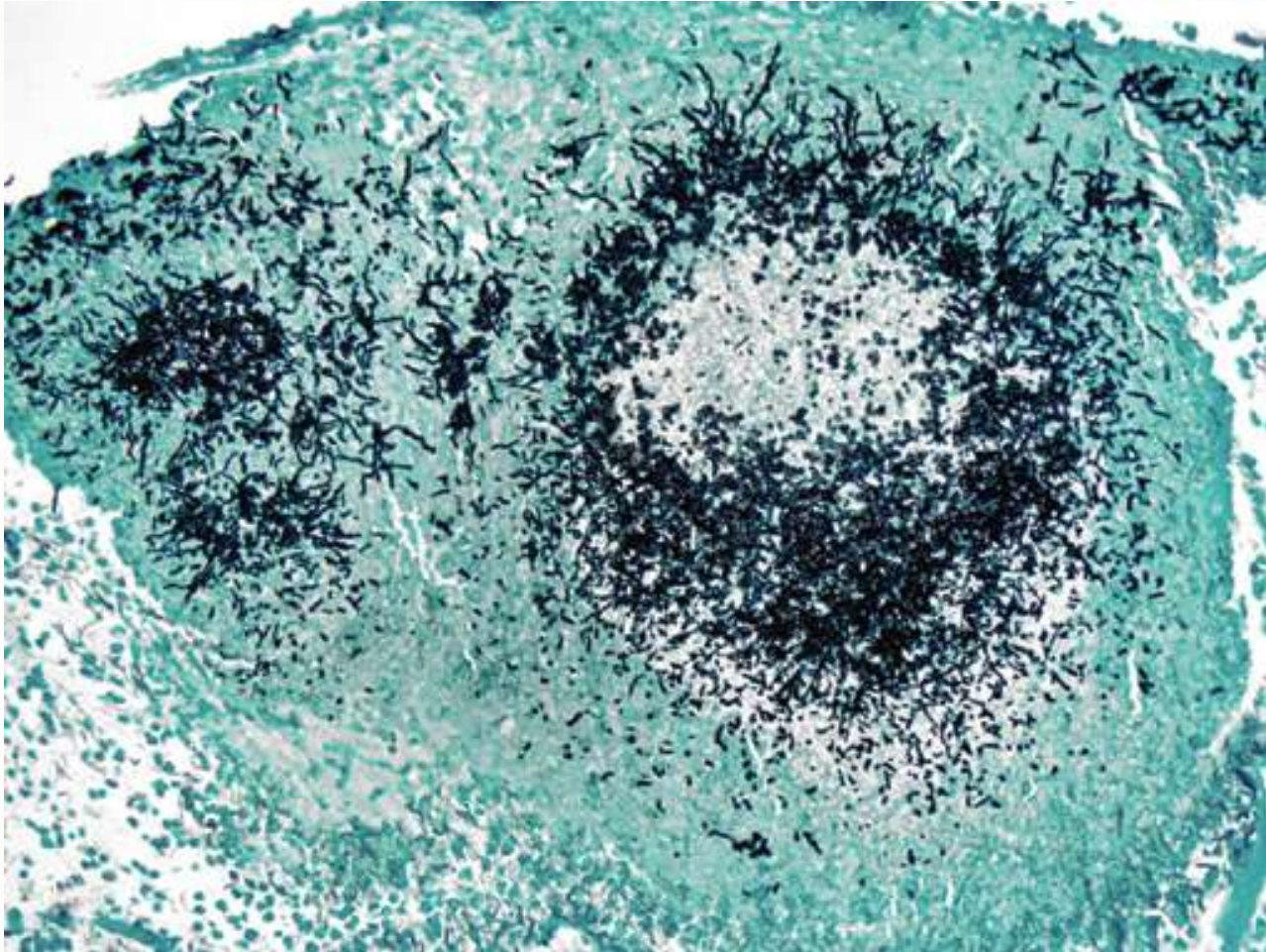
- Massive infiltrates of heterophils, multinucleate giant cells, and other leukocytes
- Granulomas had centers composed of necrotic cellular debris and heterophils with a peripheral palisade of epithelioid macrophages and aggregates of lymphocytes
- Vascular invasion of fungal hyphae causing vasculitis, thrombosis, infarction

Pathology – Microscopic



granulomatous lesion in the air sac due to aspergillosis

Pathology – Microscopic



Organisms stain well with Gomori's methenamine silver stain

Diagnosis

- Gross lesions
- Microscopic examination of scraping
- Histopathology: Special fungal stains (periodic acid-Schiff , Grocott's methenamine-silver (GMS) or calcofluor white)
- Isolation and identification of causative agent
- ELISA
- PCR



CANDIDIASIS

Introduction

- Candidiasis is caused by *Candida albicans* characterized by epithelial hyperplasia and pseudomembrane or diphtheritic membrane formation over the mucosa.
- Also known as Thrush , Crop mycosis, Sour crop, moniliasis, and Stomatitis oidica

Etiology

- *Candida albicans* is the principal agent
- Dimorphic yeast that appears as round to oval 3–4 μm budding yeasts (blastospores) on epithelial surfaces, or 3–5 μm diameter branching septate hyphae or pseudohyphae in deeper tissues.
- The condition is associated with concurrent disease, nutritional deficiency, immunosuppression or altered microflora, and is more probably the result of an opportunistic than a primary infection

Etiology

- One of the most common predisposing factors is prolonged antibiotic administration, which suppresses normal bacterial flora and competition for nutrients, thus allowing *Candida* to proliferate

Host

- Chickens, turkeys, geese, pigeons, guinea fowl, pheasants, quail, and several other avian species
- Disease is more common in birds under 3 weeks of age

Transmission

- Candida is acquired by ingestion

Pathogenesis

- Candida enter in to the body and probably becomes part of the resident flora of the mouth, oesophagus and crop → under predisposing conditions it proliferates on the surface → hyphae or pseudohyphae invade superficial epithelial layers → This invasion stimulates epithelial hyperplasia and pseudomembrane or diphtheritic membrane formation

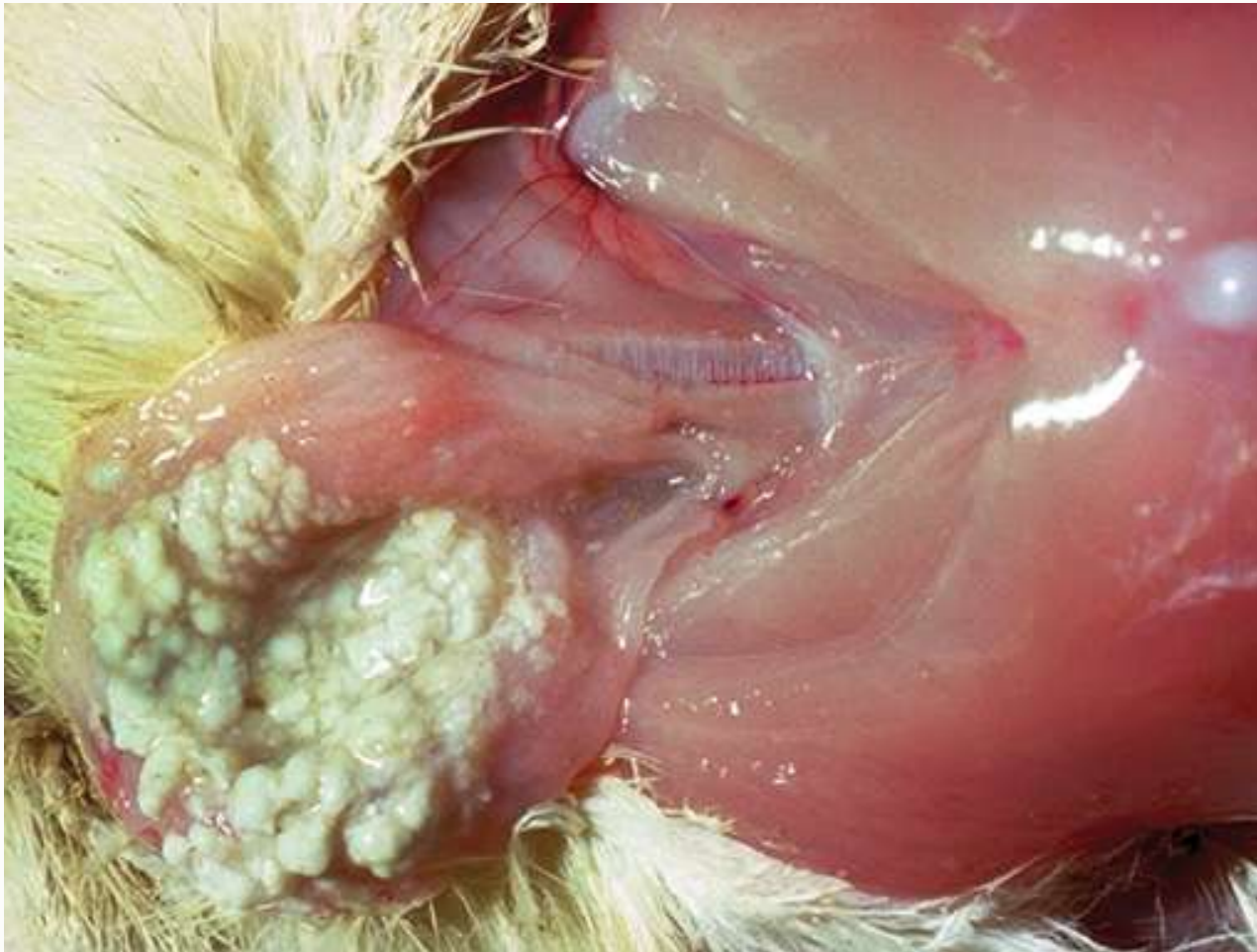
Clinical Signs

- Signs are not specific for candidiasis
- Mortality is normally not occur or very low
- Affected chicks show poor growth, stunting, listlessness, and roughness of feathers
- Signs of primary disease is predominate

Pathology – Macroscopic

- The surface of the crop and, less frequently, the oesophagus and pharynx are coated with multifocal or confluent pseudo-membrane of white cheesy material
- Pseudo-membrane has a curd-like appearance

Pathology – Macroscopic



Candidiasis (crop mycosis). Crop is markedly thickened by a soft, yellow-white pseudomembrane that has a curd-like appearance

Pathology – Macroscopic

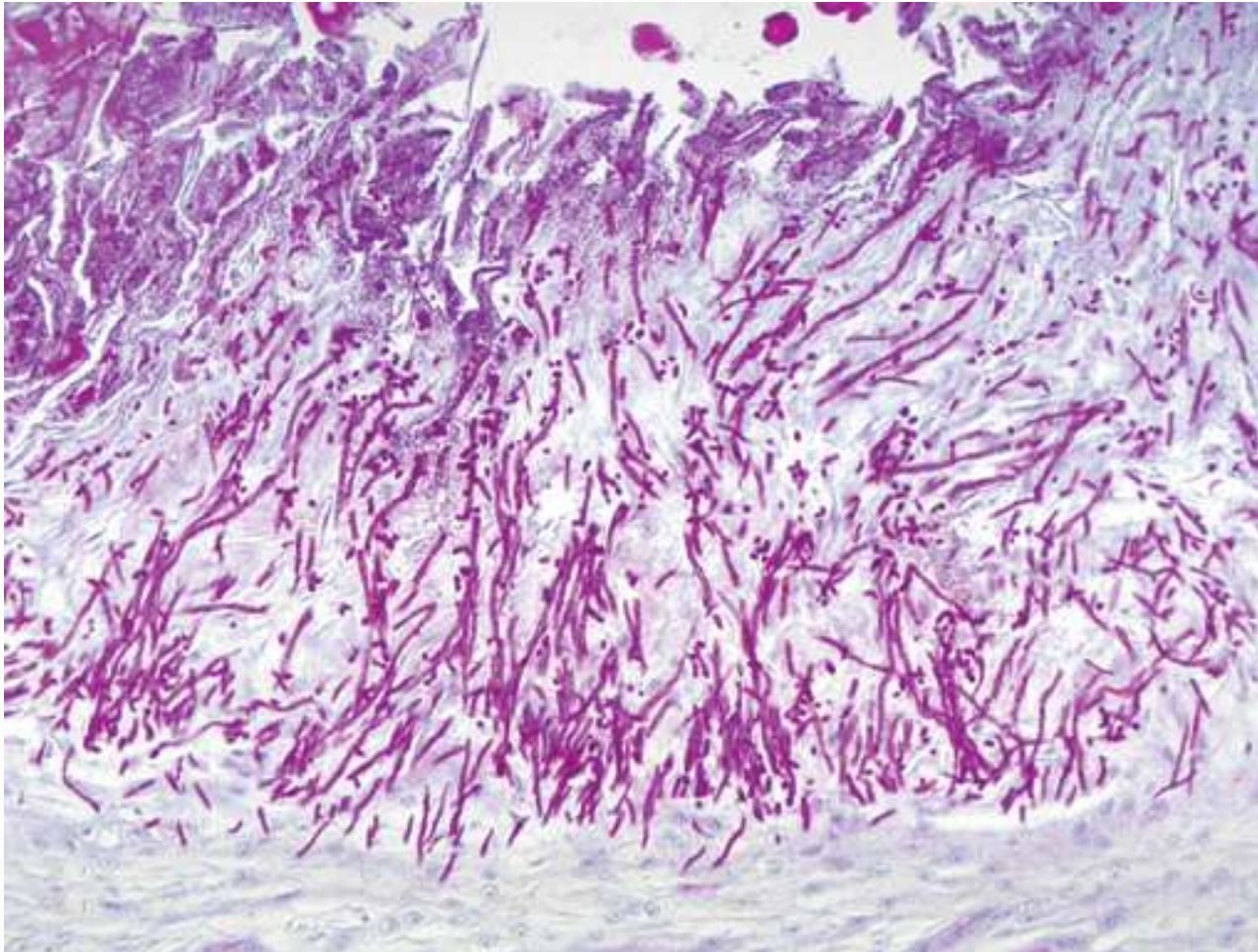


Candidiasis (crop mycosis). Crop is markedly thickened by a soft, yellow-white pseudomembrane that has a curd-like appearance

Pathology – Microscopic

- The mucosal surface is covered by exudate composed of necrotic debris, sloughed epithelial cells, leukocytes, bacterial colonies, and the yeast and pseudohyphal forms of *Candida*.
- Inflammation of epithelium with infiltration of inflammatory cells
- Dimorphic yeast that appears as round to oval 3–4 μm budding yeasts (blastospores) on epithelial surfaces, or 3–5 μm diameter branching septate hyphae or pseudohyphae in deeper tissues

Pathology – Microscopic



the perpendicular arrangement of hyphae, minimal inflammation, and lack of invasion of deeper tissues

Diagnosis

- Gross lesions
- Microscopic examination of scraping
- Histopathology: Special fungal stains (periodic acid-Schiff , Grocott's methenamine-silver (GMS) or calcofluor white)
- Isolation and identification of causative agent
- ELISA
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FAVUS

Introduction

- Dermatophytosis, dermatomycosis, ringworm, white comb and favus are terms applied to fungal infections of skin
- The infection is contagious and transmissible to humans, similar to dermatophytoses of other animals
- Infections are superficial, chronic and either self-limiting or slowly progressive
- The primary etiologic agent of favus is *Microsporum gallinae*

Introduction

- Lesions are produced initially in unfeathered skin (comb, wattle, shanks) by superficial invasion of the stratum corneum by hyphae and result in epidermal hyperplasia and hyperkeratosis
- The infection typically produces white scaly or crusty lesions on the comb and skin of the head and neck with loss of feathers
- Microscopically acanthosis, hyperkeratosis and dermatitis can be seen
- Infection can be diagnosed by histological visualization of hyphae or spores in skin lesions and feather follicles, followed by culture



Mycotoxins



AFLATOXINS

AFLATOXINS

- Aflatoxin is found in corn (maize), peanuts, cottonseed, millet, sorghum and other feed grains
- Aflatoxins are highly toxic, hepatotoxic and carcinogenic mycotoxins produced by *Aspergillus flavus*, *Aspergillus parasiticus*, and *Penicillium puberulum*
- Naturally occurring aflatoxin contains aflatoxins B1, B2, G1 and G2, but aflatoxin B1 is usually in the highest concentration and is the most toxic
- In general, ducklings and turkey poults are the most sensitive to aflatoxins and quail show intermediate sensitivity, whereas chickens are the most resistant

AFLATOXINS

- After ingestion, aflatoxin B1 undergoes biotransformation to numerous highly reactive metabolites with diverse adverse effects on metabolism. Metabolites bind to DNA and RNA, reduce protein synthesis and decrease cell-mediated immunity and to a lesser extent humoral immunity.
- Aflatoxin is rapidly excreted in the bile and urine and does not accumulate or persist in body tissues. This perhaps explains the rapid recovery of egg production and hatchability after cessation of toxin ingestion.

AFLATOXINS

- Aflatoxicosis does not usually induce mortality directly, although high levels (>10 ppm) may be lethal.
- Aflatoxin level >0.5 ppm → decrease in resistance to infections such as salmonellosis, coccidiosis, infectious bursal disease and candidiasis
- Aflatoxin level >1 ppm → decreased growth and poor feed conversion
- Aflatoxin level >2 ppm → decreased egg production and the hatchability

AFLATOXINS

- Lesions include enlarged livers which become friable and yellow with increasing dose, kidney and spleen enlargement, and reduction of the bursa of Fabricius, thymus and testes. Petechial haemorrhages or bruises after trauma are also increased because of decreased clotting factor synthesis and increased capillary fragility
- Liver has congestion, necrosis, fatty change, karyomegaly, numerous mitotic figures, bile duct hyperplasia, fibrosis, etc.
- Turkey X disease due to aflatoxin B1 + cyclopiazonic acid

AFLATOXINS





OCHRATOXICOSIS

Ochratoxicosis

- Ochratoxicosis occurs less frequently in poultry than aflatoxicosis but is more lethal because of its acute toxicity
- Ochratoxins are nephrotoxic
- Produced chiefly by *Penicillium viridicatum* and *Aspergillus ochraceus*
- Ochratoxins are isocoumarin compounds linked to L-b-phenylalanine and are designated A, B, C, and D.
- Ochratoxin A is the most common and most toxic, and is relatively stable

Ochratoxicosis

- Ochratoxin A inhibits protein synthesis, produces acute proximal tubular epithelial necrosis in the kidneys and inhibits normal renal uric acid secretion
- Affected kidneys are white to tan, swollen, hard and may have white pinpoint urate crystals and may leads to visceral gout formation
- Ochratoxin can cause glycogen storage disease in chickens

TRICHOTHECENE MYCOTOXICOSIS

- Produced by many species of *Fusarium*, *Stachybotrys*
- There are approximately 80 chemically related trichothecene mycotoxins described, but most is known about the effects of the 12,13-epoxytrichothecenes T-2, hydroxy T-2 (HT-2), diacetoxyscirpenol (DAS, anguidine) and deoxynivalenol (DON, vomitoxin)
- Trichothecene mycotoxins of significance in poultry produce their pathogenic effects by two mechanisms: direct epithelial necrosis and radiomimetic effects that destroy rapidly dividing cells

TRICHOTHECENE MYCOTOXICOSIS

- Ulcers are found at the commissures of the mouth, on the hard palate adjacent to the beak and the palatine cleft, and on the dorsal surface of the tongue
- There may be reduction in size of the bursa of Fabricius and thymus glands of young birds and birds may show anaemia and pale bone marrow

TRICHOTHECENE MYCOTOXICOSIS



Multifocal ulcerative stomatitis and glossitis

CITRININ

- Citrinin is a nephrotoxin produced in a variety of cereal grains by *Penicillium citrinum*.
- During citrinin toxicosis, mortality is rare but water consumption is increased and there is diffuse polyuria manifested as wet droppings (>300 ppm).
- Clinical signs begin within hours after initial exposure, persist as long as toxin-containing feed is consumed and stop 8–10 h after cessation of exposure.
- Grossly kidneys are slightly swollen but there are no microscopic lesions except at very high dosages.

OOSPOREIN

- Oosporein is a nephrotoxic mycotoxin produced by *Chaetomium trilaterale*.
- Ingestion by chicks (>300 ppm) or turkeys (>500 ppm) produces acute proximal tubular nephrosis and acute renal failure in severe cases.
- Effects are particularly severe in poultry less than 1 week old and lead to visceral and articular urate deposits and mortality up to 20%

Others mycotoxins

- **Fumonisin:** poultry are relatively resistant, hepatic necrosis, biliary hyperplasia, widening of growth plates, myodegeneration, etc.
- **Moniliformin:** Cardiomegaly in broilers, myocardial degeneration, hepatic necrosis
- **Fusarochromanone:** tibial dyschondroplasia
- **Cyclopiazonic acid:** skeletal muscle, heart, liver, GI degeneration
- **Zearalenone:** mycotoxin with estrogen activity; Cystic oviduct with inflammation, cysts in vent, etc.
- **Ergotism:** beak and toe necrosis; Ergotism is caused by the alkaloid toxins produced by *Claviceps* species, particularly *Claviceps purpurea* ('ergot of rye').

Thank you